INCH-POUND

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DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, MARK 45 MOD 0

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 45 Mod 0 impulse cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

2240773 - Cartridge, Impulse, Mark 45 Mod 0, Assembly

2406487 - Primer Mix No. 487

2519737 - Test Set, Closed Bomb

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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838AS208 - Propellant, Double Base

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.1.

3.2 Design and construction. The Mark 45 Mod 0 impulse cartridge covered by this specification shall be manufactured in accordance with Drawing 2240773.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Government design activity prior to manufacture (see 6.2).

3.4 Primary components. For the purpose of this specification, the primer mix (Drawing 2406487) and propellant charge (Drawing 838AS208) are considered primary components. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

3.4.1 Primer mix. The primer mix shall be compounded in accordance with Drawing 2406487.

3.4.1.1 Loading of igniter composition. The techniques of loading the igniter composition shall be such as to avoid damage to the bridgewire. The composition shall be smoothed off flush with the top of the cavity. All igniter composition shall be removed from exterior lateral areas of the spacer. Allow to dry thoroughly before cementing mylar disc in place. The dried composition shall be free from cracks, voids, and separation from the ignition cup. The igniter composition shall entirely cover the bridgewire.

3.4.2 Propellant charge (smokeless powder). The propellant charge (smokeless powder) shall be manufactured in accordance with Drawing 838AS208.

3.4.2.1 Quantity. All propellant charges shall be in accordance with Drawing 2240773. Charges shall be determined by weight rather than by volume measurement.

3.4.2.2 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.3.2.1).

3.4.2.3 Smokeless powder, handling and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.5 Insulation resistance. Prior to installation of the bridgewire, the insulation resistance between pins and case shall not be less than 2 megohms at a relative humidity of less than 80 percent when measured with a 500-volt megohmmeter (see 4.3.1).

3.6 Bridgewire. The bridgewire shall be welded to the pins.

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3.7 Bridgewire continuity. Before filling the cavity around the bridges with the igniter compound, the bridge assemblies shall be thoroughly cleaned and inspected for proper attachment and absence of projecting ends. After cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of both bridgewires. The current shall be regulated to produce a visible red glow of the bridges. An electromagnetic force of approximately 1.5 volts AC or DC may be used with a suitable series resistor to adapt the source characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both bridges shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridges.

3.8 Cartridge production. Each production cartridge shall meet the requirements of 3.8.1 through 3.8.5.

3.8.1 Insulation resistance. Prior to assembly of the bridgewire in the body and spacer assembly, the insulation resistance shall be checked with a 500-volt megohmmeter.

3.8.2 Visual. Cartridges shall be free of the following defects when inspected as specified in 4.3.2: illegible, missing, or inaccurate identification markings, incomplete threads, perforated closure disc, incomplete or improper welding of closure disc, bent pins, damaged O-rings, deep scratches, dents, sharp edges, burrs, or other defects which may prevent entry of the cartridge into the firing chamber or affect the resistance of the cartridge to moisture. Each cartridge shall meet the requirements of 3.4, 3.10, and Drawing 2240773.

3.8.3 Radiographic requirement. When radiographically inspected as specified in 4.3.3, each cartridge shall show proper assembly, presence of parts, proper welding, sealing and proper electrical connections, proper thread form, presence of the propellant, no foreign materials, and proper assembly of pyrotechnic/non-metallic materials.

3.8.4 Bridge circuit resistance. The resistance of each bridgewire in each completely assembled cartridge, when measured electrically (see 4.3.4), shall be not less than 0.130 ohm nor greater than 0.160 ohm.

3.8.5 Leakage. Each assembled cartridge shall pass a dry gas leak test. Cartridges shall not exhibit a leak rate in excess of 1×10^{-5} cm^3 per second of air at a pressure differential of 1±0.1 atmosphere (see 4.3.5).

3.9 Cartridge performance. Cartridges shall meet all the requirements of 3.9.1 through 3.9.4.

3.9.1 Shock. The cartridge shall not incur damage or deformation when subjected to the shock test specified in 4.3.6.

3.9.2 Vibration. The cartridge shall not incur damage or deformation when subjected to the vibration test specified in 4.3.7.

3.9.3 Temperature and humidity cycling. The cartridge shall not incur damage or deformation when subjected to the temperature and humidity cycling specified in 4.3.8.

3.9.4 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.9.4.1 through 3.9.4.3 when subjected to the test of 4.3.9.

3.9.4.1 Maximum pressure. The upper (U) and lower (L) limits of maximum pressure shall be U = 2300 psi and L = 1800 psi, respectively.

3.9.4.2 Misfire. There shall be no misfires (see 4.3.9.1).

3.9.4.3 Hangfire. There shall be no hangfires (see 4.3.9.2).

3.10 Workmanship. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, improper crimp, perforated closure disc, bent pins, or other defects which may prevent entry of the cartridge into the firing chamber, or affect resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.3.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Production inspection (see 4.2.2).

c. Lot acceptance inspection (see 4.2.3)

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4.2 Inspections. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.3.

4.2.1 First article inspection. First article inspection shall be performed on a sample of 64 cartridges when a first article sample is required (see 3.1). This inspection shall include the examinations and tests specified in table I. Sixty of these cartridges shall be expended in the tests listed in table I, and four cartridges shall be retained for investigative purposes. Any damage inflicted by the environmental treatments that would adversely affect the performance of the item in the service application shall be cause for rejection of the first article sample. Any further production prior to notification by the contracting agency of first article acceptance shall be at the contractor's risk. Failure of any cartridge to comply with the requirements of table I shall be cause for rejection of the first article represented. .

TABLE I. First article inspection.

Tests	Requirement paragraph	Test paragraph	A	B	C	D	E	F
Insulation resistance	3.8.1	4.3.1	12	12	12	6	6	12
Visual inspection	3.8.2	4.3.2	12	12	12	6	6	12
Radiographic inspection	3.8.3	4.3.3	12	12	12	6	6	12
Bridge circuit resistance	3.8.4	4.3.4	12	12	12	6	6	12
Leakage	3.8.5	4.3.5	12	12	12	6	6	12
Shock	3.9.1	4.3.6				6		
Vibration	3.9.2	4.3.7					6	
Temperature and humidity cycling	3.9.3	4.3.8						12
Radiographic inspection	3.8.3	4.3.3				6	6	12
Bridge circuit resistance	3.8.4	4.3.4				6	6	12
Leakage	3.8.5	4.3.5				6	6	12
Ballistic test -65 °F 70 °F 160 °F	3.9.4	4.3.9	 12	 12	 12	 6	 6	 12

4.2.2 Production inspection. All production cartridges shall be inspected and screened for the defects specified in table II. Cartridges failing to meet the requirements specified in table II, 3.2, and 3.3 shall be rejected and removed from the lot.

TABLE II. Production inspection.

Item	Tests	Requirement paragraph	Test paragraph
Body and pin assembly	Insulation resistance	3.8.1	4.3.1
Cartridge	Visual inspection	3.8.2	4.3.2
Cartridge	Radiographic inspection	3.8.3	4.3.3
Cartridge	Bridge circuit resistance	3.8.4	4.3.4
Cartridge	Leakage	3.8.5	4.3.5

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4.2.3 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in table III. Failure of any sample cartridge to comply with the requirements listed in table III shall be cause for rejection of the lot represented. Packaging defects shall be corrected before acceptance.

TABLE III. Lot acceptance inspection.

Tests	Requirement paragraph	Test paragraph	Quantity
Visual inspection	3.8.2	4.3.2	Test and retain sample
Radiographic inspection	3.8.3	4.3.3	Test and retain sample
Bridge circuit resistance	3.8.4	4.3.4	Test and retain sample
Ballistic	3.9.4	4.3.9	Test sample

4.2.3.1 Sample size. Unless otherwise specified (see 6.2), a random sample from each production lot, including the retained sample for investigative purposes, shall be selected in accordance with table IV.

TABLE IV. Lot acceptance sampling.

Lot size	Sample size	Retained sample
66 to 180	15	3
181 to 300	20	3
301 to 500	25	3
501 to 800	30	3
801 to 1,300	35	3
1,301 to 3,200	40	3
3,201 to 8,000	50	3
8,001 to 22,000	75	3

4.3 Tests.

4.3.1 Insulation resistance. Prior to assembly of the bridgewire in the body and spacer assembly, the insulation resistance shall be checked with a 500-volt megohmmeter. Units with a resistance of less than 2 megohms shall be cause for rejection.

4.3.2 Visual inspections.

4.3.2.1 Primary component inspection. The primer mix shall be visually inspected during compounding and loading for conformance to 3.4.1. The propellant charge shall be inspected for conformance to 3.4.2.2. Evidence of decomposition shall be cause for rejection. The charge quantity shall be measured for conformance to 3.4.2.1.

4.3.2.2 Cartridge inspection. The external condition and appearance of the cartridges shall be determined by comparison to Drawing 2240773. Specific inspections of the electrical connector, electrical shielding cap, threads, stainless steel case, output closure, weld, and identification label shall be conducted as a minimum. Each cartridge shall meet the requirements of 3.8.

4.3.3 Radiographic examination. All cartridges shall be radiographically examined in accordance with ASTM E1742/1742M. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridges shall be arranged on trays or boards in consecutive numerical order with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification

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shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Cartridges having any observable imperfections in assembly shall be cause for rejection. Defective cartridges shall be clearly identified on the X-ray and those defective cartridges removed from the production lot.

4.3.4 Bridge circuit resistance. Conformance with the bridgewire circuit resistance of 3.8.4 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument that is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirement of 3.8.4 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.3.5 Leakage. Each assembled cartridge shall be tested in a dry gas leak tester. This is an in-process inspection and requires 100 percent inspection of the lot. Each cartridge shall meet the requirements specified in 3.8.5. Failure to comply with the requirements of 3.8.5 shall be cause for rejection of the cartridge.

4.3.6 Shock. The shock pulse shall be applied to the cartridge's mounting points in both directions along each of three mutually perpendicular axes, for a total of 18 shocks. This test shall be conducted in accordance with method 516, procedure I of MIL-STD-810. The shock pulse wave form shall be terminal peak sawtooth. The peak amplitude shall be 15g, and the duration shall be 11 milliseconds. After being subjected to the shock test, the cartridges shall be test fired at 70 °F. Failure of any cartridge to comply with the requirements of 3.9.1 shall be cause for rejection of the lot or first article.

4.3.7 Vibration. The cartridge shall be subjected to vibration testing in accordance with MIL-D-21625. After vibration testing has been completed, the cartridges shall be test fired at 70 °F. Failure of any cartridge to comply with the requirements of 3.9.2 shall be cause for rejection of the lot or first article.

4.3.8 Temperature and humidity cycling. The test shall be conducted in accordance with the temperature and humidity cycling test of MIL-D-21625. After temperature and humidity cycling testing has been completed, the cartridges shall be test fired at 70 °F. Failure of any cartridge to comply with the requirements of 3.9.3 shall be cause for rejection of the lot or first article.

4.3.9 Ballistic test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for 24 hours at 70±5 °F and fired in a closed cylindrical chamber in accordance with Drawing 2519737. Firing current shall be not less than 7 amperes per each bridge. The cartridges shall meet the requirements of 3.9.4.

4.3.9.1 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.3.9.2 Hangfire. One or more hangfires (see 6.4.1) shall be cause for rejection of the lot represented.

4.3.9.3 Retest. There shall be no retests.

4.3.9.4 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, the original test results should be discarded and that portion of the test repeated.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mark 45 Mod 0 impulse cartridge is used in the motor control valve system of AQM-37D target drones. This impulse cartridge is only used in military aircraft, and there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Date and place of delivery.

c. When first article inspection is required (see 3.1 and 6.3).

d. Inspection conditions, if other than as specified (see 4.2).

e. Sampling, if other than as specified (see 4.2.3.1).

f. When X-rays of cartridges are required to be provided to the design activity (see 4.3.3).

g. Packaging requirements (see 5.1).

h. That the safety precaution requirements of DoDM 4145.26, Contractor's Safety Manual for Ammunition and Explosives, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of NAVSEA OP 5, Ammunition and Explosives Safety Ashore, are applicable.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Hangfire. A hangfire is defined as a firing in which the time from application of current to case opening or first indication of pressure exceeds 0.075 second.

6.5 Subject term (key word) listing.

Target drone

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – OS
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